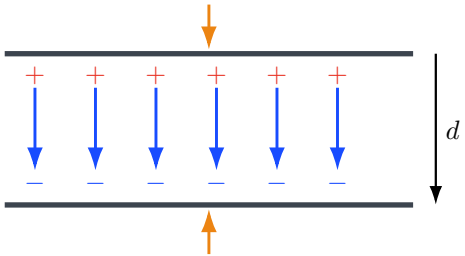
平行板电容器：固定电荷与固定电势的能量函数不同，吸力相同



两板相互吸引

$$C = \varepsilon_0 A/d, \qquad U_Q = Q^2 d/(2\varepsilon_0 A)$$

$$F_d\big|_Q = -\frac{\partial U_Q}{\partial d} = -\frac{Q^2}{2\varepsilon_0 A}$$

$$F_d\big|_V = \frac{1}{2}V^2\frac{dC}{dd} = -\frac{\varepsilon_0 A V^2}{2d^2}$$